

Anchor-Based Multimodal Verification: A Dynamic Query Framework for Fake News Forensics in Short Videos

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Abstract—The proliferation of maliciously altered short videos on social media platforms poses a significant threat to information security ecosystems, eroding public trust in digital media. Despite recent advancements in detecting fake video news, significant challenges remain in the forensic analysis of short videos, leading to issues of bias. First, as technology rapidly advances, fake videos are becoming increasingly semantically convincing, undermining the effectiveness of current classification methods. Second, the heterogeneous nature of video modalities (visual, textual, audio) creates critical challenges for models to learn discriminative feature representations. To address these challenges, we propose a dynamic query framework for fake news forensics in short videos, termed the Semantic Guided Adaptive Network (SGAN). Our approach is motivated by the need to utilize superficial alignment to identify suspicious manipulations through anchor-based verification and to leverage the adaptive capability of learnable queries to learn the heterogeneous boundary in each modality. Specifically, SGAN comprises a verification module

and a flexible query learning module. The verification module employs text as the anchor to verify detailed context, mining fine-grained information while emphasizing key features, thereby providing candidate manipulations for downstream modules. The query learning module leverages learnable queries to map heterogeneous forensic features and integrates them through multi-level fusion for decision-making. Extensive experiments conducted on two widely used datasets demonstrate the effectiveness and generalization of the proposed method. The code is released on github <https://github.com/VILAN-Lab/SGAN>

Index Terms—Fake news detection, fake news forensics, video understanding.

I. INTRODUCTION

SHORT videos are a form of conveying content with a video less than 5 minutes [1], becoming an important way for people to spread and accept information. However, the rise of maliciously crafted fake videos poses severe security risks, as individuals and entities exploit this format to manipulate public perception, propagate misinformation for political gains, and undermine trust in online content. Different from fake news videos on traditional media (e.g., YouTube), short videos benefit from the advancement and convenience of editing tools, leading to a content production method that is dominated by users.

Fake news detection on short videos [2] is a task that learns the features of video content, user comments, and metadata to determine whether a video contains misinformation. Different from previous instances of fake news that had semantic mismatches, such as altered titles, tampered voices, and modified key texts, current fake videos have mostly achieved alignment at the surface semantic level [3]. To deceive more people, they create fake videos using editing tools, computer generation, and even artificial intelligence models, making them appear authentic. The main challenge is to identify the differences between the content of real and fake videos, including both the semantic level and the low-level signal level, in order to determine the authenticity of the video.

To solve the issue of fake news videos being nearly indistinguishable from real video content, researchers [4], [5], [6], [7] have devoted great effort to analyze the misinformation in videos. Qi et al. [4] utilize the information generated by active users in short videos, e.g., user comments, likes, and metadata, to enhance the difference between fakes and reals. From the view of the relationship, Qi et al. [6] consider that the video

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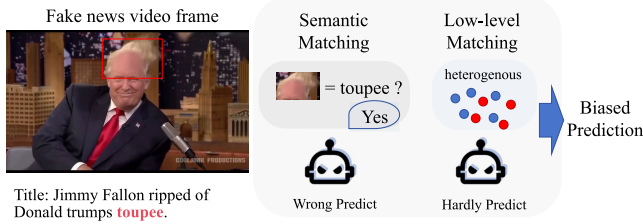


Fig. 1. The videos are related with the topic about “*Trump toupee*”. The “*toupee*” can be semantic aligned with the image area, but it is the main edited target in fake news video. The low-level matching denotes the implicit feature in computing, which is challenge for models due to the heterogeneity in various modalities. Both of them increase the biased issue in current methods.

relation is different in fake and real videos. They focus on the neighborhood relationships of news videos on the same event to distinguish similar videos. However, the above methods rely on additional information, such as comments and relationships, which means that the detection only takes effect after the video is released, rather than during the censorship. To meet the need for content moderation, Bu et al. [5] analyze the authenticity of video content from the creator’s perspective, utilizing material selection and editing to capture the unique production characteristics of fake news videos.

Despite their efforts to identify semantic differences across modalities, they are faced with the challenge that the fake news video is logically consistent at the semantic level. Some methods [8], [9] focus on low-level details, but the video encompasses various heterogeneous modalities, which also poses a challenge for the detection task, especially when metadata is missing during the censorship period. As shown in Fig. 1, the video shows Trump’s hair is a toupee, while the title also contains *toupee*. The fake content from the video and the text remains consistent. The model has difficulty detecting the fake trace at the semantic level by roughly matching. In addition, the target edited area is in the visual modality, while the text cannot directly help models detect abnormal messages in vision due to heterogeneity. Furthermore, existing models [4], [10], [11] simply fuse the multi-modal information by transformers to generate the final decision, without considering the heterogeneous expression in different modalities, aggravating the difficulty in modeling a balanced framework. As a result, the current method demonstrates biased performances in this task, as shown in Fig. 2. The differences between precision and recall show that the classifier is biased, suggesting that models are unable to balance classification. The unbalanced issue also indicates that current methods intend to categorize fake videos as real and vice versa, showing biased performance.

To address the aforementioned issues, we propose a balanced and flexible framework, called Semantic Guided Adaptive Network (SGAN), for detecting fake news on short videos, which takes into account both the semantic mismatch and heterogeneous characteristics learning. Cf. Fig. 1, even though the toupee is aligned with the image area, it also provides a direction for further analysis at the low level. The core idea of SGAN is to learn the suspicious target by independently verifying with anchor modalities and then using adaptive queries to learn the implicit fake features in

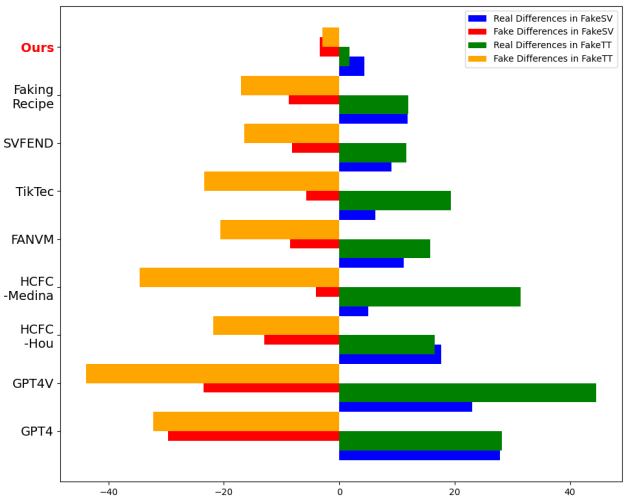


Fig. 2. The score gap between the recall and precision metrics on FakeSV and FakeTT. The orange and red bars represent the recognition score difference for false categories, while the green and blue bars represent the difference for true categories.

high dimensions. Specifically, SGAN consists of two main modules, verification and flexible query learning. The first module verifies contents based on the text modality, providing conditional fine-grained features for downstream modules, rather than directly taking matching results to final detection. Our method streamlines multimodal matching computation by aggregating modal information one-to-one based on an anchor modality, which not only slashes the complexity of interaction relationships (e.g., from 10 to 4 for 4 modalities) but also avoids the intricate task of assigning modal weights. The latter takes learnable queries to deal with various diversities and maintains the pivotal characteristics within each modality, avoiding biases from heterogeneous expressions. We also proposed a method that is conducted on the multi-channel level to enhance the low-level signal information fusion.

In this case, SGAN can preserve the intrinsic modality features, contributing to the enhancement of the discriminative embedding space, thus improving classification accuracy and balance. Unlike previously employing the raw modality representation directly, our method uses anchor-based verification as the vital information selection to avoid superficial semantic alignment. Additionally, the learnable query is applied to model the characteristic features between real and fake videos, providing a flexible framework for heterogeneous information learning. Thus, our method can handle the heterogeneous information challenge in different modalities. Our SGAN is thus capable of building a balanced classifier that learns fine-grained features based on the anchor modality and utilizes learnable queries to fit the multi-modal learning in flexible. Experiments on two widely used datasets demonstrate the superiority and generalizability of our SGAN. The visualization results show the explicit features learned by our model.

Our main contributions are summarized as follows,

- To overcome the unbalanced challenge, we present a balanced and flexible framework for fake news detection on short videos, which learns the fine-grained distinguished

features according to the anchor modality and utilizes learnable queries to learn multi-modal features for fusion.

- The verification module takes the superficial semantic alignment as auxiliary information to learn suspicious features for learnable queries, avoiding the harmful effect of authentic-looking videos. The flexible queries learning flexibly deals with the heterogeneous modal information by learnable queries and incorporating multi-modal features in different levels.
- Extensive experiments on various datasets consistently validate that our approach can achieve new state-of-the-art fake news detection on short video results on main metrics. The additional experiments show that our method is balanced and generalizable.

II. RELATED WORK

A. Fake News Video Detection

The detection of fake short videos can be roughly divided into two categories based on the status of video publication. The first one is the analysis and detection of video content by platforms during censorship. In the early stage, the handcraft feature-based methods are widely used in fake news video detection. The models extract the speech, emotion, and video title vectors for SVM models to learn the decision boundary [8], [9]. Currently, for pre-publication detection, researchers analyze the content of the video at the data signal level based on the text to determine the presence of misinformation [5]. Zeng et al. [12] address the issue of modality missing in the detection process by leveraging domain knowledge enhancement and contrastive learning method. Some researchers implement pre-detection by detecting watermarks in videos. Fu et al. [13] use a wavelet-domain cascaded encoder network to address the issues of low visual quality and insufficient robustness of existing watermarking methods in screen capture scenarios. Some methods transform information from other modalities into one specific modality, which can be considered anchor-based methods. For instance, it transforms visual information into textual information through transcription (e.g., CA-FVD [14]) before verifying and computing the textual modality information, while our method does not require such transformation, thereby reducing information loss during the conversion process.

The other approach is the detection by the platform after publication, combining more information such as comments and relation [4], [6], [7], [15], [16], [17], [18]. For videos that have already been published, the model can analyze the authenticity of the video by integrating more information. Qi et al. [4] use information such as the number of likes, user profiles, and comment content to determine the authenticity of the video. To make use of the debunking video, Qi et al. [6] exploit the neighborhood relationship between fake and debunked videos to enhance the model's effectiveness. Furthermore, Zong et al. [7] mine the implicit opinion in short videos via the large language model and take the diffusion method to simulate the propagation of opinions. Zeng et al. [19] mimic the human reasoning process relying on evidence and achieve fine-grained fake news detection through

the comprehensive analysis of news content and corresponding evidence. Even though these methods exhibit impressive representation abilities, they may struggle to distinguish the fake or real videos precisely due to the same editing and complex semantics. Our proposed SGAN solves the biased classifier by leveraging the adaptive framework to alleviate the heterogeneous challenge. This design is based on the characteristic that false information in fake short videos may exist across multiple modalities, which avoids the adverse impact of videos that appear authentic.

B. Video Classification

Video classification [20] is a fundamental task that classifies videos into class labels, and it has many applications in fields such as deepfake classification [21], [22], action recognition [23], and multimedia analysis [24], [25]. Early studies on video classification directly employ hand-crafted features to construct classification models [26], [27]. Due to their reliance on human experience and domain knowledge, these methods fail to advance rapidly. Recently, deep learning methods have achieved remarkable results in video classification. They employ the long short-term memory network (LSTM) [28], [29] and convolutional neural networks (CNN) [30], [31] to learn features automatically. Specifically, they take CNNs to encode spatial features (e.g., visual features in one frame) [32] and adopt LSTMs to encode temporal features (e.g., frames in the video) [33]. Due to the emergence of transformers, which have a stronger ability to encode temporal information, a large number of methods for video classification tasks have adopted methods based on transformers [34], [35], [36]. Researchers use transformers as a multi-modal encoder, leveraging the attention mechanism to learn video information [37], [38], [39]. Deepfake video detection [21] is also an important direction in video detection. Liao et al. [22] leverage the Dempster-Shafer theory to fuse predictions, addressing the problem of compression artifacts such as blurring. However, these methods focus on the content presented in videos while neglecting the assessment of the authenticity of the video information. Unlike them, our method focuses on misinformation detection by learnable queries to enhance the model's robustness.

III. METHODOLOGY

A. Problem Formulation

The fake news detection on short videos can be formulated as follows: given a set of videos represented by four modalities, i.e., text T , vision V , audio A , and emotion E , the goal of the task is to classify the video into real or fake with the given information. Emotion is the key means to express the author's intent in videos. Different emotional tendencies in audio can elicit varying viewing experiences for the audience. Through sentiment analysis, it can be ensured that the video content aligns with the expectations of the target audience, thereby emotion can be one of the main content for reasoning.

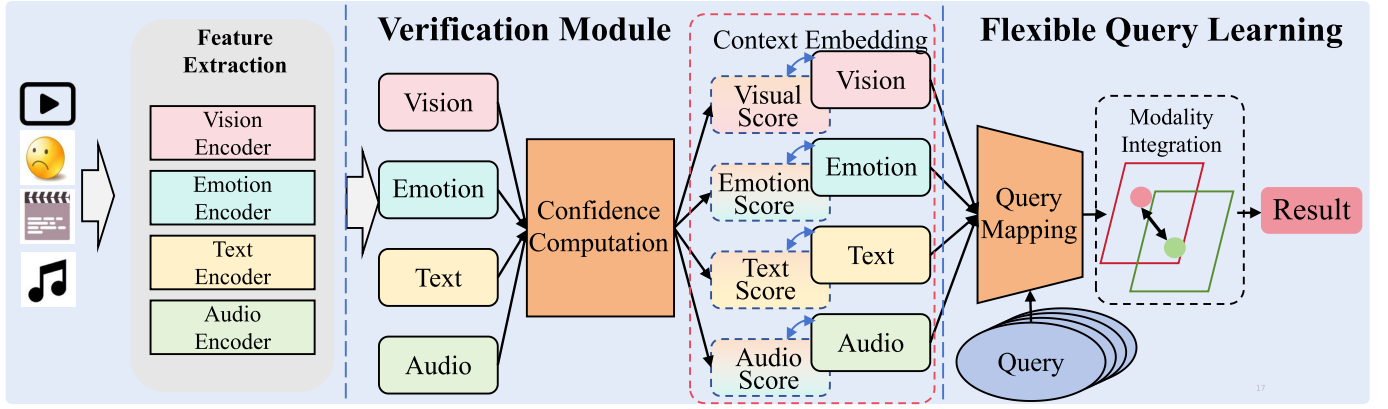


Fig. 3. The architecture of our SGAN. The verification module contains two sub modules, i.e., confidence computation and context embedding, while the flexible query learning contains the query mapping and modality integration.

B. Proposed Method

As shown in Fig. 3, our model verifies suspicious targets in each modality feature with anchor information, then uses learnable queries to deal with the vagaries of situations, so SGAN can handle fake news short videos in a flexible way. In particular, given the above modality features as input, our method first takes the text feature as the anchor to compute the confidence score and then updates the context features. Subsequently, SGAN leverages learnable vectors as queries to focus on fake traces, obtaining the characteristic feature in each modality. These features are aggregated via the integration network with each other in multi-aspect for alignment and fusion. The aggregated feature is fed into the feed-forward network to achieve classification. We will provide a detailed explanation of our method in the following.

1) *Features Extraction & Encoding*: To extract text features, we utilize CLIP [40] to encode titles & transcripts to token level semantic features $T = [t_1, t_2, \dots, t_n]$, $t_i \in \mathbb{R}^{D_1}$, where t_i and D_1 denote the i -th token feature and the dimension, respectively. For the vision feature, we first leverage TransNetv2 [41] to segment for video clips and then also take CLIP to obtain frame semantic features $V = [v_1, v_2, \dots, v_m]$, $v_i \in \mathbb{R}^{D_2}$, where m denotes the max frame number. For the audio feature, we leverage the pretrained wav2vec [42] to extract the representation $A \in \mathbb{R}^{D_3}$. We employ OpenSmile [43] for the extraction of audio emotion features $E \in \mathbb{R}^{D_4}$. To encode the processed features, we utilize a linear layer to project them into a common semantic space as:

$$f_* = \text{ReLU}(\text{Linear}(f_*^{\text{raw}})), \quad (1)$$

where $f_* \in \{f_t, f_v, f_a, f_e\}$ and f_t, f_v, f_a, f_e denote the text, video, audio, and emotion features, respectively.

2) *Verification Module*: Previous methods for detecting fake videos primarily involved checking for semantic consistency to identify news, such as examining the alignment between text and vision. However, due to fake videos are becoming increasingly realistic, applying this method to current fake news videos would easily result in misclassification.

To resolve the issue, we propose the verification method to encode modality features with anchor, which checks the

differences between modalities like a form of prior judgment based on the text. Our method is also capable of reducing the number of matching calculations between multiple modalities.

Since the text semantic feature f_t is considered as the anchor, the other modality features, i.e., vision, audio, and emotion, are the compared targets. We choose the textual modality as the anchor modality based on two core considerations. First, the textual modality has inherent advantages of direct information expression and low noise. Second, it references the prior focus on the textual modality in previous works. In addition, to enhance text feature effectiveness, it also compares with itself. We choose the textual modality as the anchor modality based on two core considerations. First, the textual modality has inherent advantages of direct information expression and low noise. Second, it references the prior focus on the textual modality in previous works [4], [14], [44]. For the sake of brevity, we mainly introduce the computation process between text and visual modalities.

As shown in the left box of Fig. 4, to learn confidence scores between visual frames and text, the model first computes the conditional feature f_c as:

$$f_c = \frac{((f_v + f_{pos}) \cdot W_{Q1}) \cdot (f_t \cdot W_{K1})^T}{\sqrt{d_k}} \cdot (f_t \cdot W_{V1}). \quad (2)$$

Here, the query consists of the visual features f_v augmented with position embedding (PE) f_{pos} , while the key and value are represented by the text features f_t themselves. The d_k is the projection channel dimension. W_{Q1} , W_{K1} , and W_{V1} are the learnable weight matrices for query, key, and value, respectively. Then, to compute the confidence score S , we adopt a Gaussian kernel function to project the text features and visual feature to high-dimensional space, making them linearly separable. The more similar the two features are, the closer the kernel function value is to 1. Specifically, the confidence score S is computed by normalizing text features and visual features along the taking their dot product, and then scaling and transforming the result using a temperature scaling factor as:

$$S = \alpha \cdot \exp\left(-\frac{(1 - F(f_c)^T F(f_v))^2}{2\sigma^2}\right), \quad (3)$$

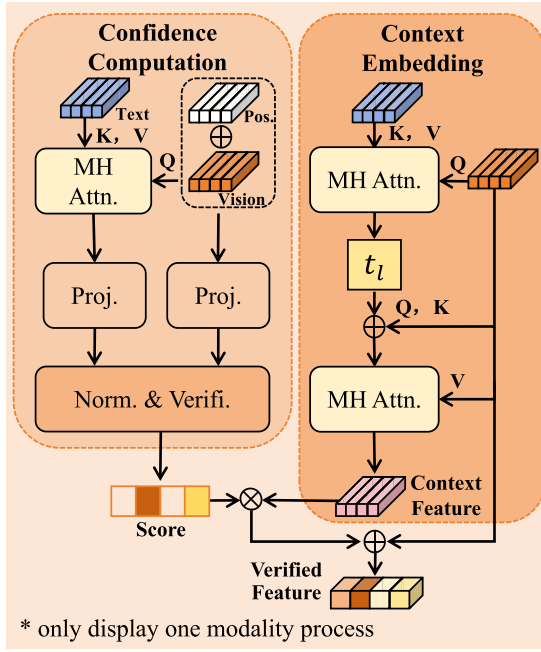


Fig. 4. The detail for the verification module. It is worth noting that when dealing with discriminative features, VER is applied to each modality, so it processes each modality in parallel separately, and maintaining weight sharing. The left box is the confidence computation, while the right is the context embedding.

where α and σ are learnable parameters, and F denotes the linear layer for projection. This score represents the credibility rating of the text for each modality, and can mitigate the representation bias caused by modal heterogeneity.

As shown in the right box of Fig. 4, together with obtaining the confidence score, the context embedding module enhances the characteristic information as:

$$t_l = \frac{(f_v \cdot W_{Q2}) \cdot (f_t \cdot W_{K2})^T}{\sqrt{d_k}} \cdot (f_t \cdot W_{V2}), \quad (4)$$

$$f_{cont} = \frac{((t_l + f_v) \cdot W_{Q3}) \cdot ((t_l + f_v) \cdot W_{K3})^T}{\sqrt{d_k}} \cdot (f_v \cdot W_{V3}). \quad (5)$$

Here, we employ the visual feature and the text feature as the query and key for encoding the text-based features t_l , which can gather important features for verified features based on the vision. Thereafter, we add t_l to visual features f_v as the query and key, while the f_v is the value for the attention mechanism to obtain context features f_{cont} . The process means that we represent the vision content in text form, also to mitigate the representation bias caused by modal heterogeneity.

To create a verified feature representation, we fuse text features and context features by summing them and then multiplying the sum by the confidence score S . The visual verified feature $\hat{f}_v$ can be defined as:

$$\hat{f}_v = f_{cont} \cdot S + f_v. \quad (6)$$

By confidence and context encoding, our method encodes modality features independently, providing the highlighted differences for downstream modules.

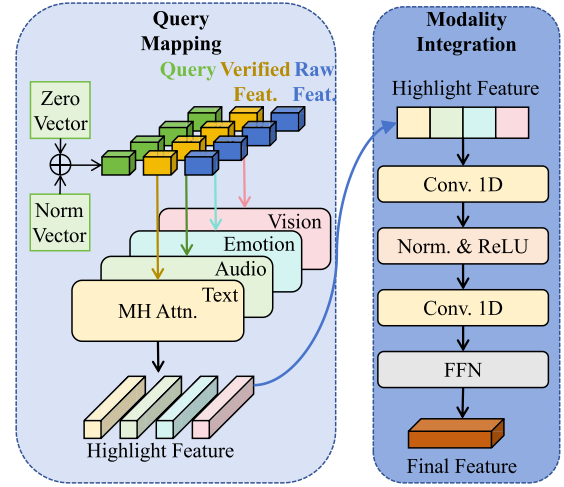


Fig. 5. The detail for the flexible query learning. The left box is the query mapping, while the right is the modality integration. The *Verified Feat.* and *Raw Feat.* denote the verified feature from each modality (e.g., $\hat{f}_v$) and raw feature (e.g., f_v), respectively.

3) *Flexible Query Learning*: The modified trace is presented in any modalities, while these features are heterogeneous and do not directly align with others, challenging for fake short video detection. To address the issue of the key information alignment, we propose a flexible query learning method, which leverages learnable queries to fit important contents and takes a multi-level convolution to achieve adaptive fusion. Specifically, this method contains two steps, i.e., query mapping and modality integration.

For brevity, we mainly introduce flexible query learning for visual modality. As shown in the left of Fig. 5, we first initialize the query embedding $q_v \in \mathbb{R}^{1 \times d}$ as the implicit characteristic in videos. Referring previous work [45], it is randomly initialized by sampling from a uniform distribution ranging from -1 to 1 . Then, to learn the implicit characteristic in videos, we also initialize a zero query $\hat{q}_v$, which has the same shape as q_v . The randomly initialized queries are used to consider the heterogeneous information of multi-modal data, while the zero-initialized queries are used as a supplement to handle the characteristic information of real and fake videos. The learnable query can flexibly adapt to implicit fake traces through parameterized learning, without relying on manually designed heuristic rules. The process can be formulated as:

$$b_v = \frac{((q_v + \hat{q}_v) \cdot W_{Q4}) \cdot (\hat{f}_v \cdot W_{K4})^T}{\sqrt{d_k}} \cdot (f_v \cdot W_{V4}). \quad (7)$$

Here, the query of the attention mechanism is the addition of the q_v and $\hat{q}_v$, while the key and value are verified features from the VER module. They are fed into multi-head attention to obtain the primary highlight feature b_v corresponding to the visual modality, ensuring that it can take into account the false information present in the vision.

Since the four category highlight features (i.e., b_t , b_v , b_e , and b_a) are modeled separately in the high dimension space, the next phase is to align and fuse them, comprehensively calculating the features of different modalities. Specifically,

the modality integration module is designed for aligning four features at the channel level and dimension level, which can utilize the complementarity between modalities to enhance robustness rather than directly take fusion. The module first performs a concatenation for query embeddings and highlight features to derive learnable features b_l :

$$b_l = [b_t + \hat{q}_t, b_v + \hat{q}_v, b_e + \hat{q}_e, b_a + \hat{q}_a], b_l \in \mathbb{R}^{4 \times D_5} \quad (8)$$

For the channel level alignment, as Eq. 9, the module uses a one-dimensional convolution with a kernel size of 1 to convolve over a four-channel vector. Its purpose is to linearly combine the channel features at each position in the input sequence, focusing on cross-channel feature fusion at the same position while keeping the sequence length unchanged. After passing through the convolutional layer, we use the layer norm (LN) and ReLU activation function to filter the information used for fusion. Different from the channel level, the feature at the dimension level has more fine-grained information for the classification. To achieve it, the module employs a transposition operation that swaps the positions of the first and second dimensions of the feature b'_l , which is denoted as Eq. 10. Then, the module also applies a 1D convolution over the input feature b'_l to derive the aligned feature b''_l . To keep the learnable query effective, the aligned feature and learnable query are then concatenated in the modality dimension (i.e., the dimension 4) and merge it and feature dimension (i.e., D_5) into a single dimension $4D_5$, to convert the 3D tensor into a 2D tensor. Finally, we fed it into a feed-forward network to compute the predicted score:

$$b'_l = \text{ReLU}(\text{LN}(\text{Conv}(b_l))), b'_l \in \mathbb{R}^{4 \times D_5} \quad (9)$$

$$b''_l = \text{Conv}((b'_l)^T), b''_l \in \mathbb{R}^{4 \times D_5} \quad (10)$$

$$f_f = \text{FFN}(\text{cat}[(b''_l)^T, q_i]), f_f \in \mathbb{R}^{1 \times 4D_5}, \quad (11)$$

where $\text{LN}(\cdot)$ denotes the layer normalization, $\text{FFN}(\cdot)$ is a feed-forward network composed of two linear projection layers with ReLU activations, $\text{Conv}(\cdot)$ is a 1D convolution layer, q_i is the learnable query, and f_f is the final feat for decision.

IV. EXPERIMENTS

In this section, we evaluate the proposed method on two commonly used datasets to verify the effectiveness of SGAN and the necessity of its individual components.

A. Datasets

FakeSV [4] is the largest Chinese fake news short video dataset, encompassing a wide range of fields. We conduct our experiments on the temporal split. It has 1,810 samples of fake and 1,814 samples of real videos, with the train, validation, and test set divided in a ratio of 70%, 15%, and 15%, respectively. Note that the ratio of real to fake videos in the test set is 238:304. Following [5], we focus on early detection, and train the model without using the data after publication.

FakeTT [5] is an English fake news short video dataset, which sources from TikTok for evaluating the performance of models in English-speaking contexts. It contains 1,172 fake news videos and 819 real news, and the ratio of real to fake videos in the test set is 200:99.

B. Experimental Settings

Considering our focus on content-based detection in the initial phase of news dissemination, all benchmark methods have been modified to depend exclusively on content.

We adopt the Adam optimizer to train our model with an initial learning rate of $5e-5$, and a batch size of 128. The model is trained for 30 epochs with the early stop for 5 epochs. The layers have the same dropout rate of 0.1. The attention mechanism has 8 heads, and 1D convolution layers are arranged with a kernel size of 1, stride of 1, and padding of 0. For the learnable parameters in Eq. 3, we set $\alpha = 1.0$ and $\sigma = 0.5$ as the initial values. The PE is sampling from a uniform distribution ranging from -1 to 1 . Since sentiment is non-sequential data, it does not require positional embedding, and the vector for PE is set to *None*. We set the maximal number of video frames m as 83, length of text n as 512. The dimension D_1 of FakeSV text is 768, while in FakeTT is 512. The D_2 , D_3 , D_4 , and D_5 are 512, 768, 768, and 512, respectively.

Following Bu et al. [5], the versions of GPT-4 and GPT-4V we used are GPT-4-0613 and GPT-4-vision-preview in zero-shot prompting setting, respectively. Specifically, as shown in Fig. 6, we input video descriptions and OCR-extracted text into GPT-4. For GPT-4V, the inputs include video descriptions, OCR text, and Base64-encoded images. Regarding the time cost, for inference on the FakeSV test set, GPT-4 takes approximately 8 hours, while GPT-4V takes around 20 hours. For inference on the FakeTT test set, GPT-4V takes roughly 5 hours, and another run of GPT-4V takes about 10 hours. For the InternVL2.5-8B zero setting, we adopt the same input with the GPT-4V, while the finetune version, we take the official LoRA method to train the model. All of the local experiments are conduct on a GeForce RTX A6000.

C. Evaluation Metrics

Following the previous work [4], [5], [6], we report Accuracy (Acc) and Macro F1 in percentage as primary evaluation metrics. To account for imbalanced label distributions, we additionally report the F1-score, Precision, and Recall for each category. Although the F1 score can reflect both recall and precision, when a unbalanced model has increasement on both recall and precision, the absolute difference may increases, which is not easily reflected in the F1 score. To demonstrate our model balanced performance, we compare a new metric, Matthews Correlation Coefficient (MCC) in percentage [46], which is derived from the absolute difference between recall and precision within the category. MCC is stable across varying positive-negative ratios with symmetric weights, whereas F1 score may favor the majority class in imbalanced distributions.

D. Baselines

We compare our model with the following baselines:

- **Large language models: GPT-4** [47] is an advanced AI model with high capabilities in natural language understanding. We employ the “gpt-4-0613” version for our task. **GPT-4V** [48] is a variant of the GPT-4 model, which

Input for GPT-4	
Text Prompt:	You are an experienced news video fact-checking assistant and you hold a neutral and objective stance. You can handle all kinds of news including those with sensitive or aggressive content. Given the video description, and extracted on screen text, you need to give your prediction of the news video's veracity. If it is more likely to be a fake news video, return 1; otherwise, return 0. Please refrain from providing ambiguous assessments such as undetermined.
Description:	{video description}; On-screen Text: {extracted on-screen text}; Your prediction (no need to give your analysis, return 0 or 1 only):
Input for GPT-4V	
Text Prompt:	You are an experienced news video fact-checking assistant and you hold a neutral and objective stance. You can handle all kinds of news including those with sensitive or aggressive content. Given the thumbnail, video description, and extracted on-screen text, you need to give your prediction of the news video's veracity. If it is more likely to be a fake news video, return 1; otherwise, return 0. Please refrain from providing ambiguous assessments such as undetermined.
Description:	{video description}; On-screen Text: {extracted on-screen text}; Upload Image: data:image/jpeg;base64,{thumbnail} Your prediction (no need to give your analysis, return 0 or 1 only):

Fig. 6. The prompt we used in GPT-4 / GPT-4v. The image input way for GPT-4V include passing image URLs or Base64-encoded images. We upload local images by converting them into Base64 encoding.

TABLE I

PERFORMANCE COMPARISON BETWEEN SGAN AND BASELINES ON THE FAKESV AND FAKETT DATASETS. THE BEST PERFORMANCE IN EACH COLUMN IS BOLD. THE SECOND-BEST RESULTS ARE INDICATED BY UNDERLINE. THE * DENOTES THAT THE MODEL IS FINETUNED ON THE CORRESPONDING DATASET. THE VALUES IN () ARE THE RELATIVE STANDARD DEVIATION VALUES OBTAINED AFTER TEN INDEPENDENT RUNS. THE VALUES IN [] ARE THE P-VALUES FOR OUR METHOD AND BASELINES

Dataset	Method	Acc	Macro F1	MCC	Fake			Real		
					Precision	Recall	F1	Precision	Recall	F1
FakeSV	GPT-4 (2023)	67.43 (0.0049)	67.34 (0.030) [1.31e ⁻²⁸]	41.90	83.71	53.99	65.64	57.81	85.71	69.05
	GPT-4V (2023)	69.15 (0.0039)	69.14 (0.033) [3.27e ⁻²⁷]	43.24	82.35	58.78	68.60	60.00	83.08	69.68
	InternVL2.5-8B (Arxiv'24)	68.26 (0.0043)	67.10 (0.039) [5.47e ⁻²⁷]	34.82	69.41	77.63	73.29	66.34	56.30	60.91
	InternVL2.5-8B*(Arxiv'24)	82.47 (0.0025)	81.48 (0.016) [6.88e ⁻³³]	65.13	78.79	94.08	85.76	89.94	67.65	77.22
	HCFC-Hou (ICMI'19)	74.91 (0.0049)	73.61 (0.038) [9.32e ⁻²⁵]	48.82	73.46	86.51	79.46	77.72	60.08	67.77
	HCFC-Medina (ACL'20)	76.38 (0.0037)	75.83 (0.021) [1.55e ⁻²⁷]	51.79	77.50	81.58	79.49	74.77	69.75	72.17
	FANVM (CIKM'21)	79.52 (0.0044)	78.81 (0.037) [6.51e ⁻²²]	58.25	78.64	87.17	82.68	80.98	69.75	74.94
	TikTec (BigData'21)	73.43 (0.0029)	73.26 (0.027) [4.90e ⁻²⁷]	46.75	78.37	72.70	75.43	68.08	74.37	71.08
	SVFEND (AAAI'23)	80.88 (0.0015)	80.54 (0.011) [2.12e ⁻⁵²]	60.79	<u>85.82</u>	77.63	81.52	74.53	83.61	78.81
	FakingRecipe (MM'24)	<u>85.35</u> (0.0015)	<u>84.83</u> (0.011) [1.15e ⁻⁴⁰]	<u>70.11</u>	83.33	<u>92.11</u>	<u>87.50</u>	<u>88.35</u>	76.47	<u>81.98</u>
	SVRPM (COLING'24)	79.34 (-)	78.55 (-) [-]	-	-	-	-	-	-	-
	CA-FVD (ICIC'25)	84.50 (0.0497)	84.12 (0.063) [9.28e ⁻¹⁵]	68.42	84.16	89.14	86.58	85.00	78.57	81.66
	SGAN (ours)	86.35 (0.0104)	86.06 (0.011)	72.20	86.39	89.80	88.06	86.28	81.93	84.05
FakeTT	GPT-4 (2023)	61.45 (0.0060)	60.66 (0.0071) [7.52e ⁻⁴¹]	25.45	43.36	75.61	55.11	83.19	55.00	66.22
	GPT-4V (2023)	58.69 (0.0049)	58.69 (0.0053) [1.56e ⁻⁴⁵]	33.62	44.52	<u>88.46</u>	59.23	88.00	43.42	58.15
	InternVL2.5-8B (Arxiv'24)	55.85 (0.0084)	55.82 (0.0106) [2.86e ⁻³⁸]	23.91	41.45	80.81	54.79	82.08	43.50	56.86
	InternVL2.5-8B*(Arxiv'24)	62.54 (0.0042)	62.49 (0.0070) [2.09e ⁻³⁹]	37.46	46.56	88.89	61.11	<u>90.00</u>	49.50	63.87
	HCFC-Hou (ICMI'19)	73.24 (0.0039)	72.00 (0.0039) [2.01e ⁻⁴¹]	46.54	56.93	78.79	66.10	87.04	70.50	77.90
	HCFC-Medina (ACL'20)	62.54 (0.0037)	62.23 (0.0042) [2.26e ⁻⁴⁶]	32.69	46.24	80.81	58.82	84.92	53.50	65.64
	FANVM (CIKM'21)	71.57 (0.0028)	70.21 (0.0041) [4.81e ⁻⁴²]	42.77	55.15	75.76	63.83	85.28	69.50	76.58
	TikTec (BigData'21)	66.22 (0.0018)	65.08 (0.0020) [1.08e ⁻⁶⁷]	33.64	49.32	72.73	58.78	82.35	63.00	71.39
	SVFEND (AAAI'23)	77.14 (0.0012)	75.63 (0.0012) [2.88e ⁻⁶⁷]	52.67	62.33	78.79	69.57	87.91	76.33	81.69
	FakingRecipe (MM'24)	79.15 (0.0011)	77.74 (0.0012) [8.93e ⁻⁶²]	57.02	<u>64.75</u>	81.82	72.18	89.74	<u>77.83</u>	83.30
	CA-FVD (ICIC'25)	<u>80.93</u> (0.0028)	<u>79.69</u> (0.0033) [2.78e ⁻²²]	60.85	66.67	84.85	74.67	91.33	79.00	<u>84.72</u>
	SGAN (ours)	81.93 (0.0011)	79.81 (0.0011)	<u>59.66</u>	71.84	74.75	<u>73.27</u>	87.24	85.50	86.36

further optimizes multi-modal comprehension capability. **InternVL2.5** [49] is widely-used vit-mlp-llm framework, incorporating pixel unshuffle preprocessing and dynamic high resolution processing.

- **Handcraft feature models:** **HCFC-Hou** [8] uses SVM to classify fake news videos by analyzing the speech text and acoustic emotion. **HCFC-Medina** [9] extracts TF-IDF vectors and frames features, and then employs SVM and logistic regression to learn the boundary.
- **Neural network models:** **FANVM** [15] utilizes the title and video frames to extract topic agnostic features for classification. **TikTec** [16] fuses visual object features speech information for classification via an attention

module. **SVFEND** [4] exploits the correlations between multi modalities for detection. **FakingRecipe** [5] is a dual-branch model for detection that captures useful clues from the author perspective. **SVRPM** [11] is a detection model using pretraining tasks. **CA-FVD** [14] is a model designed to detect the authenticity of news videos by matching visual, textual, and auditor modalities.

E. Result and Analysis

1) *Overall Performance:* As detailed in Table I, the experimental results show that the performance of our method is overall superior to the previous models. We conduct statistical

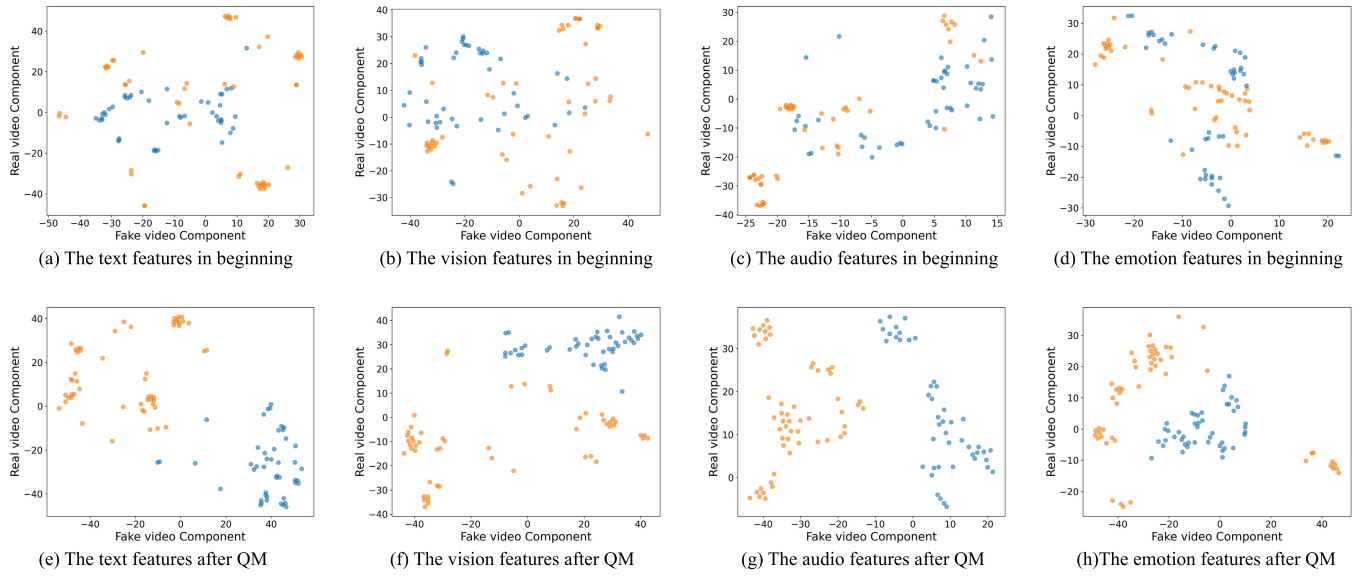


Fig. 7. The visualization of four modality features on the FakeSV test set. The blue and the yellow dots are denoted real and fake news video representations, respectively.

significance tests on the macro-F1 metrics obtained from 10 independent runs for all baseline models and our method. As shown in Table I, the result indicates that the performance improvement of SGAN over baselines is statistically significant.

Due to the well performance of LLMs in numerous tasks, we have selected them as comparative models. Despite the strong generalization capabilities of LLMs, they perform poorly in fake news video detection in the zero-shot setting, with significant biases in classification. Specifically, these models (i.e., GPT-4 & GPT-4V) tend to classify videos in the FakeSV dataset as real and videos in the FakeTT dataset as fake, especially GPT-4 in the setting of lacking visual information. On the FakeSV dataset, GPT-4v achieves an MCC score 28.96 points lower than our method, and 26.4 points lower on the FakeTT dataset. Meanwhile, on the FakeSV dataset, GPT-4 has an MCC score 30.3 points lower than our method, and 34.21 points lower on the FakeTT dataset. As for the improvement of the InternVL-8B, when complete multimodal inputs are considered, LoRA fine-tuning of the model can effectively improve its performance. However, it still achieves a low score (65.13 in FakeSV, 37.46 in FakeTT) in the MCC metric, indicating poor performance in terms of balance. The possible reasons for the overall poor performance of LLMs are as follows: 1) The model is required to process unseen domain data, which forces it to reason based on its internal knowledge. 2) Videos from different datasets vary in clarity, and FakeSV provides higher video resolution, which offers more visual details to facilitate reasoning.

After fine-tuning, LLMs, e.g., InternVL2.5-8B, have achieved performance comparable to that of our method, which fully demonstrates their potential in this field. However, training large models requires thousands of hours of GPU time [49], and the fine-tuning process also takes 5 hours on an A6000, which is far longer than our training and testing time

of only 2 minutes per epoch for 30 epochs (cf. Table X). The massive number of parameters in large models makes them difficult to deploy on edge devices. Our method not only achieves more balanced classification performance (72.20 vs. 65.13 in the MCC metric) but also enables inference and training under more stringent memory constraints. Our method focuses on the imbalance problem in fake news detection on short videos. This problem is also pronounced in large models. For example, GPT-4V tends to classify samples as fake news, leading to bad performance on the MCC metric (43.24 in FakeSV and 33.62 in FakeTT). Moreover, fine-tuning fails to resolve this issue, as the MCC score of the fine-tuned InternVL2.5-8B still shows a significant gap compared with that of our proposed method. The massive number of parameters in large models allows them to quickly adapt to tasks across various domains, yet this becomes a drawback when it comes to requirements such as debiasing and balanced classification. Our method leverages learnable queries to avoid the adverse effects of plausible videos and flexibly handle heterogeneous modal information and it is capable of achieving balanced classification in a relatively balanced semantic space (cf. Fig. 7). Exploring the integration of the rich semantic knowledge of large models with the flexible semantic space of small models constitutes an important research direction for future work.

Secondly, by referring Table I, we observe that from hand-craft feature-based baselines to neural network-based models, the performance (e.g., F1 score) is continuously improving, but there is no significant decrease on the MCC metric, which indicates that the previous works are lack of considering the balance in classification. Thirdly, SGAN outperforms all previous methods on accuracy, macro F1, and MCC in both datasets, and in terms of sub-category metrics, our model also exhibits good performance. Particularly, we achieve an improvement on the balanced metric. Our model surpasses FakingRecipe with a margin of 1.23 percentage points (p.p.)

TABLE II

ABLATION STUDY OF MULTIPLE MODEL COMPONENTS ON TEST SETS

Module				Dataset					
VER		FQL		FakeSV			FakeTT		
CC	CE	QM	MI	Acc	F1	MCC	Acc	F1	MCC
✓	✓	✓	✓	86.35	86.06	72.20	81.93	79.81	59.66
✓	✓	✓	✓	84.50	84.22	68.46	78.26	77.28	57.46
✓	✓	✓	✓	76.75	76.09	52.49	76.25	75.11	52.75
✓	✓	✓	✓	85.79	85.39	71.12	79.59	78.02	56.92
✓	✓	✓	✓	85.98	85.68	71.45	80.26	78.90	59.06
✓	✓	✓	✓	85.97	85.54	71.57	77.25	75.64	52.38
✓	✓	✓	✓	84.87	84.55	69.19	80.60	79.22	59.61

and 2.01 p.p. in Macro F1 for FakeSV and FakeTT, respectively, while SGAN outperforms FakingRecipe by 2.09 p.p. and 2.64 p.p. in MCC for the same datasets. Here, we demonstrate that SGAN can achieve the balance in classification.

2) *Ablation Study*: To further evaluate the effectiveness of the verification (abbr. VER) and flexible queries learning (abbr. FQL) modules, we conduct the ablation experiments between our models with the variants. We can see that the VER and FQL are both effective for our framework (Cf. Table II). Specifically, the VER and FQL module have a significant impact on the performance of our model on both datasets. The FQL module, responsible for calculating classification boundaries, has a greater impact on model performance than the VER module. This is evidenced by the larger drop in MCC scores when ablating FQL (−19.71 p.p.) compared to VER (−3.74 p.p.) in the FakeSV dataset. In addition, we explore the impact of components in detail. The confidence computation (abbr. CC) and context embedding (abbr. CE) still have a greater impact on FakeTT. Moreover, the confidence scores provided by the CC sub-module can help the model better integrate multimodal information. The query mapping (abbr. QM) is the core of our model and is almost equivalent to that of the entire module. The modality integration (abbr. MI) module show different effects on different datasets, which influences the FakeSV dataset more significantly. We also observe that the model variations exhibit more pronounced fluctuations in MCC metric, while demonstrating relatively minor changed performance in F1 score. We attribute this to that the MCC can better evaluate model performance on imbalanced datasets.

The results show the importance of each modality in Table III. When removing the vision modality, the performance drops the most on the Fakesv dataset, while the impact on the Fakett dataset is minor. As mentioned in the main text, we consider that the video quality is different in the two datasets, which causes varied influences on the model. Specifically, for the vision modality, removing poor-quality vision information makes the model more difficult to classify the majority class (leading to a decrease in accuracy metric), and also hinders the model to distinguish between real and fake samples, especially on imbalanced datasets. When the sentiment modality is removed during the calculation process, the decline in the model’s balanced performance on FakeTT

TABLE III

THE ABLATION STUDY OF MODALITIES ON TEST SETS. THE *T*, *V*, *E*, AND *A* DENOTE THE TEXT, VISION, EMOTION, AND AUDIO, RESPECTIVELY

Modality	Dataset					
	FakeSV			FakeTT		
T	V	E	A	Acc	F1	MCC
✓	✓	✓	✓	86.35	86.06	72.20
✓	✗	✓	✓	84.50	84.25	68.51
✓	✓	✗	✓	85.97	85.52	71.59
✓	✓	✓	✗	85.60	87.74	70.84

TABLE IV

DIFFERENT MULTI-MODAL FUSION OPERATIONS ON TEST SETS

Module	FakeSV			FakeTT		
	Acc	F1	MCC	Acc	F1	MCC
Ours	86.35	86.06	72.20	81.93	79.81	59.66
MLP	84.31	83.60	68.58	78.59	77.90	60.16
Mean	85.79	85.43	71.09	80.26	78.66	58.05
Add	85.64	85.97	71.44	76.58	75.57	54.11
Att	85.61	85.29	70.70	76.59	75.64	54.55

is the most significant among all modalities. For the decrease in removing audio, we attribute that audio contains little information since it is background music or ambient sound, which increases the difficulty of differentiation.

3) *Variant Models*: To verify the effectiveness of the proposed method in the multimodal fusion stage, we conducted comparative experiments between this method and mainstream fusion strategies. The comparison objects include fully connected fusion after feature concatenation (MLP), mean fusion (Mean), additive fusion (Add), and self-attention fusion (Att). The experimental results are shown in Table IV. The proposed method (Ours) can integrate multimodal information more efficiently and achieve better classification performance on both types of datasets. The MLP strategy is significantly lower than Ours in the three core metrics of Acc, F1, and MCC on the FakeSV dataset, indicating that its fusion capability is greatly affected by the dataset and its generalization performance is insufficient. The Mean strategy shows that the simple mean weighting method has limited adaptability to tasks with different data distributions. The attention mechanism performs poorly in multimodal fusion at the end of the model. We believe the reason for the poor performance of other fusion methods is that they only consider a single feature dimension while ignoring the multimodal fusion alignment in the sequence, and this is exactly the advantage of our method.

To verify the effectiveness of our model architecture, we have conducted experiments using different feature extractors. As shown in Table VII, our model is trained in the feature extractor settings of FakingRecipe and SVFEND, respectively. The term “fixed” indicates that frames are extracted from videos on average. In the TransNetv2 +CLIP setting, our model outperforms FakingRecipe by 1.23 p.p. (1.45%) and 2.07 p.p. (2.67%) in Macro F1 on the FakeSV and FakeTT datasets, respectively. In the Fixed+VGG setting, our model achieves a 3.56 p.p. (4.42%) higher Macro F1 score than

TABLE V
THE EXPERIMENT ABOUT MISSING MODALITIES

Missing modality	FakeSV		FakeTT	
	F1	MCC	F1	MCC
Text	77.31	54.80	79.50	59.03
Emotion	85.83	71.83	76.80	54.26
Vision	80.96	62.02	79.26	59.01
Audio	85.54	71.10	79.69	59.46

TABLE VI
EXPERIMENTS WITH DIFFERENT MODALITY FEATURE ANCHORS

Anchor	FakeSV			FakeTT		
	Acc	F1	MCC	Acc	F1	MCC
Text (Ours)	86.35	86.06	72.20	81.93	79.81	59.66
Vision	86.34	85.69	72.99	78.92	77.39	55.82
Emotion	84.87	84.55	69.19	79.26	78.02	57.80
Audio	85.79	85.45	71.08	81.60	79.77	59.80

TABLE VII
EXPERIMENTS WITH DIFFERENT FEATURE EXTRACTOR. THE MODEL NAMES IN PARENTHESES () INDICATE THAT OUR MODEL IS TRAINED IN THE CORRESPONDING MODEL SETTING

Variant	FakeSV		FakeTT	
	Acc	F1	Acc	F1
TransNetv2+CLIP (FakingRecipe)	86.35	86.06	81.93	79.81
Fixed+CLIP (-)	84.50	84.25	80.60	78.52
Fixed+VGG (SVFEND)	84.32	84.10	80.27	78.20
TransNetv2+VGG (-)	84.50	84.27	80.94	78.94

SVFEND on FakeSV and a 2.57 p.p. (3.40%) higher one on FakeTT. To further explore this, we experimented with different backbone combinations, namely Fixed+CLIP and TransNetv2 +VGG. We found that the performance of our model remains highly competitive compared to existing methods in these settings.

4) *Model Cost*: To demonstrate the computational cost incurred by the sentiment modality, we compared our method with three models. One is the variant of our method without the sentiment modality, and two are baseline models, namely SVFEND [4] and FakingRecipe [5]. In this experiment, we keep the same settings for all models, where the batch size is set to 128, video frames is 83, image size is 448×448 , and tool is the thop library¹ with Pytorch 2.7.1 version. The FLOP calculation covers the process from input to gradient return. Since we aim to preserve the original structures of each model as much as possible, we do not change the SVFEND model, which incorporates the BERT model for word vector processing and the last two layer of the VGGish model for audio feature processing, while the FakingRecipe model leverages the processed feature. Our model, SGAN, keeps the same setting with the FakingRecipe model. As shown in Table X, our method occupies a relatively large amount of memory, which is attributed to the VER module that contains multimodal interaction information and generates large-sized

TABLE VIII
DIFFERENT MULTI-MODAL FUSION OPERATIONS ON TEST SETS

Method	FakeSV			FakeTT		
	Acc	F1	MCC	Acc	F1	MCC
SVFEND	80.88	80.54	58.25	77.14	75.63	52.67
(+VER)	83.94	83.61	67.30	80.26	78.82	58.71
TikTec	73.43	73.26	46.75	66.22	65.08	33.64
(+VER)	82.47	81.84	64.41	80.27	79.12	60.14

TABLE IX
HASHTAG COUNT STATISTICS OF DATASETS

Dataset	Sample Type	Avg. Hashtags
FakeSV	REAL	0.95
FakeSV	FAKE	0.99
FakeTT	REAL	7.18
FakeTT	FAKE	7.9

TABLE X
COMPARISON OF PERFORMANCE AND RESOURCE CONSUMPTION ACROSS MODELS

Model	FLOPs	Parameter	Memory	Train Time
SGAN (ours)	29.65 G	1.58 M	14362 M	2 min/epoch
SGAN w/o emo	29.56 G	1.25 M	13968 M	2 min/epoch
FakingRecipe	59.00 G	12.82 M	4226 M	6 min/epoch
SVFEND	840.10 G	224.07 M	6541 M	8 min/epoch

feature maps. Notably, incorporating the sentiment modality increases the memory usage by less than 10%, indicating that the sentiment module has a negligible impact on the overall model. However, although our method's memory footprint is relatively larger than that of the comparative models (approximately 2–3 times), its number of parameters is only one-tenth of theirs (Ours 1.58M vs. SVFEND 224.07 M). Additionally, integrating the sentiment modality does not significantly increase the computational cost of our method, and our method uses the fewest FLOPs and parameters among all compared methods. The high FLOPs of the FakingRecipe method are primarily due to the dominant matrix multiplication operations in its large-dimensional MLP, while the SVFEND model embeds a pre-trained audio processing module into its network structure, leading to increased FLOPs and parameters.

5) *Generalizability*: For end-to-end models, the VER module is a plug-and-play method. To assess the generalizability of VER, we adopt the TikTec [16] and SVFEND [4] with VER module. As shown in Table VIII, we observe that incorporating VER resulted in performance gains on both baselines, with TikTec showing significant improvements. We consider that the VER module improves the capability of finding suspicious targets for reasoning, and making inferences more easily.

In real scenarios, the text in user-generated videos may be missing or of low quality. Therefore, we conduct experiments about the fully trained model in the modality missing setting. Unlike Table III, the experiments in Table V involve training the model on the complete dataset and then testing it with different modalities absent. As shown in Table V, the result shows that our method is robust in real-world scenarios. From the results, the lack of textual information has the greatest

¹<https://pypi.org/project/thop/>

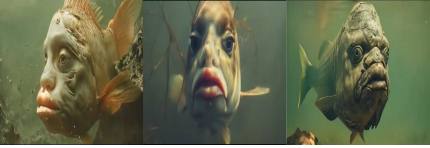
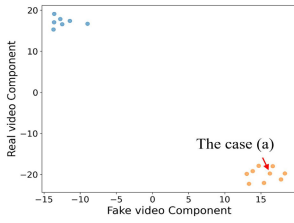
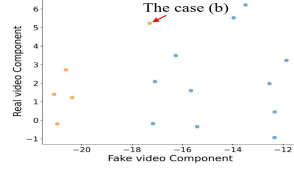
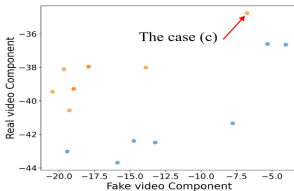
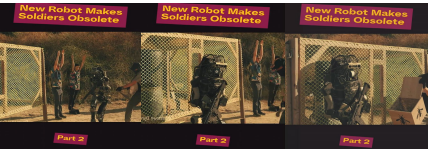
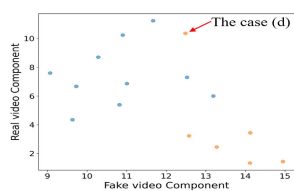
Input	Result	Feature Visualization	Web-derived Explanations
(a)  Title: Have you ever heard of the human faced fish? The Homo Piscis is said to be found at Lake Samsara in AFRICA #homopiscis #lakesamsara #fish #humanfacedfish	SV-FEND: Fake ✓ FakingRecipe: Fake ✓ CA-FVD: Fake ✓ Ours: Fake ✓		The video is AI-generated, and has low quality in resolution.
(b)  Title: beirut explosion was caused by missile dont be fooled.	SV-FEND: True ✗ FakingRecipe: True ✗ CA-FVD: Fake ✓ Ours: Fake ✓		The case is a video that the missile was edited into the original video, and a filter was added to give the illusion of thermal or infrared imaging.
(c)  Title: 鸵鸟：以后叫我滑雪小能手 OCR: The ostrich skis better than national-level athletes.	SV-FEND: True ✗ FakingRecipe: True ✗ CA-FVD: True ✗ Ours: Fake ✓		The title means that <i>Ostrich: From now on, call me a skiing expert!</i> It is a 2003 advertisement by JR East Japan Company.
(d)  Title: New Robot Makes Soldiers Obsolete Part 2 #marines #training #specialforces #usamarines #military #soliders	SV-FEND: True ✗ FakingRecipe: True ✗ CA-FVD: Fake ✗ Ours: True ✗		The error case, which is a video that is used CGI to Fake Military Robots Shot on Location California Tactical Academy.

Fig. 8. Case (a), (b), and (d) are from the FakeTT dataset, while Case (c) is from the FakeSV dataset. The explanations are collected by the author, rather than appearing in the dataset. The blue and the yellow dots are denoted real and fake news video representations, respectively.

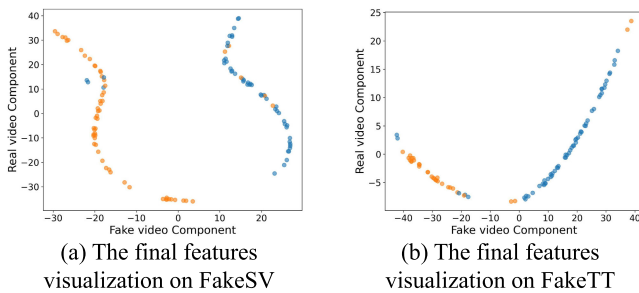


Fig. 9. The visualization of final decision features on the FakeSV test set. The blue and the yellow dots are denoted real and fake news video representations, respectively.

impact on the model on the FakeTT, followed by visuals, audio, and sentiment. On the FakeSV, the greatest reduction is caused by sentiment. We attribute this to the different effects of emotional expression in different languages, with English videos exhibiting emotions more prominently.

To further analyze the impact of different modalities on model performance, we visualized the modal features of the two datasets and statistically analyzed their linguistic distribution characteristics. As shown in Fig. 10, in the FakeTT dataset, the emotional modality (red) and textual modality (green) are close to each other in distance after clustering in

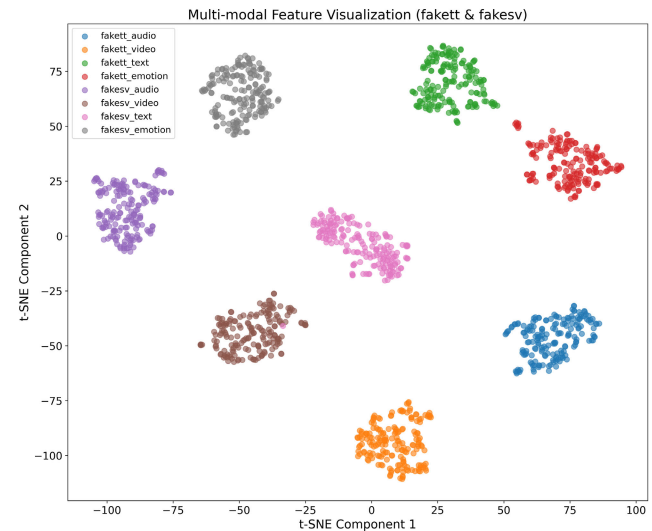


Fig. 10. The visualization of features on the FakeSV and FakeTT dataset.

the feature space. This indicates a strong correlation between the textual and emotional modalities. When the emotional modality is missing, it will weaken the effect of the textual features relying on the emotional features for reasoning.

To address the significant impact of textual information absence on the FakeSV dataset, we observed and statistically

analyzed the two datasets. The text in the FakeSV dataset mostly consists of sentences that summarize video content, while the text in the FakeTT dataset is mostly tags such as “#real”. Based on this observation, we conducted statistics on the tags in the datasets. As shown in Table IX, the average number of “#” symbols in the FakeSV dataset is 1, whereas that in the FakeTT dataset is 7. This indicates that the descriptive summary sentences provided by the FakeSV dataset play an important role in model performance. In contrast, the tag information in the FakeTT dataset is too limited and contains additional tags, resulting in more noise. Thus, removing it does not overly affect model performance.

6) *Confirmatory Experiment*: To confirm the prior assumption, that is “the text has sufficient information to meet the conditions of the anchor feature”, we conduct comparative experiments by replacing different modalities as anchors, and the results are shown in Table VI. As the results, the textual modality achieves the optimal performance when used as the anchor. The performance of other modalities is correlated with their characteristics to the dataset. For example, when the visual modality is used as the anchor, its performance is significantly affected by video quality. On the FakeSV dataset with high-definition videos, its MCC metric even outperforms that of the textual modality. On the FakeTT dataset, however, the model’s balance capability drops sharply. When the emotion is used as the anchor, it cannot provide a stable benchmark for verifying other modalities due to the lack of rich semantic information, resulting in poor overall performance. When the audio serves as the anchor, its information volume is limited, mainly consisting of background music or ambient sounds. However, it can provide scene attributes (such as environmental information) to establish verification connections with the textual, visual, and emotional modalities, so its performance is close to that of the textual anchor. In summary, our choice of the textual modality as the anchor primarily hinges on its dual advantages of low noise and rich semantics. Meanwhile, the experiments also confirm that our method exhibits good adaptability to the anchor modality. Even when other modalities are selected as the anchor, the method still achieves decent performance, which demonstrates its flexibility across different scenarios.

F. Visualization

To show the feature distribution learned by our modules, we visualize each modality representation after query mapping using the t-sne [50]. To ensure category balance, we randomly select 50 samples from real and fake videos, respectively. Cf. Fig. 7 (from a to d), we observe that the features of real videos are mixed with those of fake videos, especially with a high degree of overlap in the text and vision modalities. After the query mapping, we can see that the features of real and fake videos have already formed fairly distinct boundaries. There are still some points within the range of the other category since fake videos are similar to reals. We also show the visualization of final features for decisions in Fig. 9. In general, the yellow and purple dots are mostly clearly separated. However, at symmetrical positions, such as the inflection point in Fig. 9(a) and the two ends in Fig. 9(b),

the yellow and purple dots are mixed together. We believe that such occurrences may be due to the minimal differences between real and fake videos. We will provide such cases for analysis in case study. Therefore, we can demonstrate in a visual manner that our model is capable of achieving a good balance in classification.

G. Case Study

We present four cases to demonstrate our method’s effectiveness. To analyze the reasons for classification, we present the visualization of the sample’s visual features within its respective batch in the third column. As shown in Fig. 8(a), the video² exhibits obvious fictional characteristics, such as a fish with a human face. In addition, the video has low quality in resolution. From the visualization, the features of the two classes (real and fake) form concentrated clusters through clustering in this sample batch. In this case, all models can predict correctly. Cf. Fig. 8(b), the video³ is edited to cause panic by adding a missile. To enhance credibility, the text also says *caused by missile dont be fooled*, which has consistency in text and vision modality. The Case (b) lies outside the cluster of the fake class, indicating that it is difficult to classify this sample using only visual features. Our model is not deceived by this superficial consistency. On the contrary, it is able to further analyze the missiles in the images, thereby making the correct decisions. The Case (c)⁴ is sourced from a Chinese-language platform. It claims that the ostrich has been trained and its skiing skills surpass those of professional athletes. The ostrich depicted skiing is actually a synthesis of professional skiers dressed in ostrich costumes and real ostriches filmed in a studio. The visualization also demonstrates that the visual features of this fake video are close to those of real videos. Videos with similar image-text content lead the CA-FVD model, which relies on image-text consistency, to make false judgments. Our model makes determinations by analyzing the synthesized texture details. For the Case (d), it is generated through CG and meticulous manual adjustments,⁵ and these videos maintain a high degree of consistency at the signal level. This kind of fake video is easily confused with real video, which is also the reason why there is some mixing in our final classification results. For instance, all models make incorrect predictions in Fig. 8 (d).

V. CONCLUSION

In this paper, to solve the challenge of the unbalance rising from superficial alignment and heterogeneous gap, we propose a semantic guided adaptive network (SGAN) for fake news detection on short videos. Concretely, to obtain the suspicious target in each modality, SGAN takes the anchor-based verification and context embedding to retain the integrity of the information while highlighting the key features. To deal with

²The factcheck detail: <https://averagebeing.com/fact-check-is-the-human-faced-homo-piscis-fish-real-or-fake/>

³The factcheck detail: <https://factcheck.afp.com/doctored-video-falsely-claims-show-missile-striking-beirut-port>

⁴The factcheck detail: <https://www.snopes.com/fact-check/skiing-ostriches-real/>

⁵The factcheck detail: <https://www.youtube.com/watch?v=zOyDwBvxAyA>

heterogeneous expressions, our method utilizes the learnable query to capture the implicit features adaptively and incorporates multi-modal features at different levels. Experiments show that our method achieves a comparable performance with the state-of-the-art models on two widely used datasets. The additional experiments demonstrate our SGAN has generalizability and balance.

As the quality of AI-generated videos continues to improve significantly, traditional methods of detecting synthetic content through video quality analysis or simply multi-modal consistency checks are proving increasingly inadequate. Our work highlights the importance of leveraging semantic guidance and adaptive learning in the context of fake news forensics, offering a robust solution to the evolving challenges in digital media authentication.

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